Mathematics - Undergraduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The polynomial x^4 + 4 can be factored into linear factors in Z_5[x]. Find this factorization.
   (a) (x - 2)(x + 2)(x - 1)(x + 1)
   (b) $(x+1)^4$
   (c) $(x-1)(x+1)^3$
   (d) $(x-1)^3(x+1)$
2. How many structurally distinct Abelian groups have order 72?
   (a) 4
   (b) 6
   (c) 8
   (d) 9
3. A tree is a connected graph with no cycles. How many nonisomorphic trees with 5 vertices exist?
   (a) 1
   (b) 2
   (c) 3
   (d) 4
4. If a real number x is chosen at random in the interval [0, 3] and a real number y is chosen at random in the interval [0, 4], what is the probability that $x < y$?
   (a) 1/2
   (b) 7/12
   (c) 5/8
   (d) 2/3
5. In xyz-space, what are the coordinates of the point on the plane 2 x + y + 3 z = 3 that is closest to the origin?
   (a) (0, 0, 1)
   (b) (3/7, 3/14, 9/14)
   (c) (7/15, 8/15, 1/15)
   (d) (5/6, 1/3, 1/3)

True / False (10 marks)

Answer True or False

6. True or False: If H is a normal subgroup of G, then aH = Ha for all a in G.
7. True or False: (g o $h)^2 = g^2$ o h^2 for every g, h in an abelian group G.
8. True or False: The cosets of the subgroup 4 Z in 2 Z are 4 Z and 2 + 4 Z.
9. True or False: The value of f(8) for the function f(x) defined by f(2 n) = n^2 + f[2(n - 1)] is 36.
10. True or False: There are exactly two abelian groups of order 4 up to isomorphism.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In right triangle ABC with $\angle B = 90^\circ$, we have $\sin A = 2\cos A$. What is $\tan A$?
12. Discuss the implications of an eigenvector v of an invertible matrix A. Explain why v must also be an eigenvector for the matrices 2 A, A^2, and $A^($-1).

End of Paper